

TRANSPORTED ANTI-SPOOFING THRESHOLDS FAIL CONSERVATIVELY: MISSED SPOOFS AT A FALSE-ALARM RATE BELOW TARGET

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ABSTRACT

A speech-deepfake detector’s threshold, set to 5% false-positive rate (FPR, bona fide rejected) on one channel, rarely holds it on another. Across 108 deployment cells built from the real transmissions of ASVspoof 2021 LA and four corpora, 77 realize an FPR more than $2\times$ off target (36 above, 41 below). Of the 72 cells inside our own ± 5 -point tolerance on realized FPR, 41 are among them, and every one is conservative: the tolerance reads clean exactly where spoofs pass. In the costliest resolvable cell, AASIST calibrated on a PSTN transmission and deployed on the same recordings over G.722 realizes **0.42% FPR** while missing **70% of spoofs, against 0.66% at the deployment-calibrated threshold**. The failure persists on ASVspoof 5 when calibration and deployment share neither recordings nor speakers: 187 of 264 ordered pairs (71%) miss by more than $2\times$ across two detectors. A target-channel bona-fide order statistic restores *marginal expected* FPR control: at $N=500$, mean realized FPR is 4.91–5.06% over twelve detector–corpus cells, with at most a 0.5-point FNR premium over the deployment oracle wherever a usable operating point exists. The guarantee is marginal, not per deployment: its Beta dispersion law assumes iid sampling, and speaker clustering widens it on every tested cell. A label-free importance-weighted quantile does not restore control, and no tested unlabeled monitor survives an attack-prevalence shift. Threshold transport can therefore fail quietly; target-channel calibration restores marginal control, and its sampling unit and dispersion must be reported.

Index Terms— Anti-spoofing, audio deepfake detection, threshold calibration, conformal prediction, domain shift

1. INTRODUCTION

Audio anti-spoofing studies usually report performance at the EER threshold, an oracle operating point selected with labels from both evaluation classes. A deployed service has no such oracle: it must choose a threshold in advance and accept the resulting false-positive rate. This difference can hide severe deployment failures. A threshold set to 5% FPR on ASVspoof 2019 LA dev [1] realizes 22–98.5% FPR across

21LA/21DF [2], In-the-Wild [3], and BRSpeech-DF [4] for a strong open-source detector [5]; for XLS-R+SLS [6], it reaches 99.5%. These results agree with the transfer failures in [7] and the bona-fide-driven cross-testing errors in [8]. Channel is a major source of variation: [9] finds that presentation can matter more than model scale, and a recent challenge targets this problem directly [10]. Language introduces a complementary resource asymmetry. LRLspoo [11] transfers an EER threshold across 66 languages but, without target-domain bona fide, can report only spoof rejection. Our deployment regime assumes the converse resource: target bona fide is available, but target spoof labels are not. We therefore evaluate the FPR side.

We study the deployment behavior of a standard statistical repair: given N labeled target-domain *bona-fide* recordings and no spoof labels, the $\lfloor (N+1)\alpha \rfloor$ -th order statistic has marginal rank validity under exchangeability without ties, and under iid continuous sampling its conditional FPR follows an exact finite-sample Beta law [12, 13]; conservative Neyman–Pearson variants give high-probability upper bounds [12, 14]. Neither the statistic nor bona-fide-only audio calibration is new [15]. Our contribution is empirical: we measure transport across channels and corpora, quantify its operational cost, replicate the failure on disjoint data, and compare finite-sample policies at matched labeling budgets, asking whether target-channel calibration leaves a *usable* operating point and where its sampling assumptions fail.

We make three contributions (details in §§4–5). (1) The measurement covers 108 deployment cells: two detectors $\times$ (42 ordered pairs of seven 21LA channel conditions + 12 ordered pairs of four corpora). We report both the loss of false-alarm control and the spoof-side cost relative to each deployment’s oracle. (2) We re-audit the bona-fide quantile against parametric and unlabeled alternatives at matched resource. (3) We report negative results: the tested label-free heuristic does not restore control, and no tested unlabeled drift monitor survives an attack-prevalence shift. We claim neither the threshold rule nor the practice of auditing both error rates at a fixed operating point; both are prior art [12, 13, 16, 17]. Our two detectors lose control by different amounts, but they

differ in both front-end and static accuracy, so $n=2$ cannot separate the explanations.

2. RELATED WORK

Order-statistic calibration is established: Neyman–Pearson methods control type-I error through class-0 order statistics [12, 14] and negative-only conformal calibration provides marginal guarantees [13]. In audio, CA-SOADD calibrates a bona-fide-only α -quantile [15], Reality Check fixes thresholds before evaluation [18], [7] audits threshold transfer and seven unlabeled corrections, and [8] documents bona-fide-driven cross-testing errors. Relative to that audit’s one detector and two target corpora, we add a 108-cell fixed-FPR map that includes real 21LA transmissions, a finite- N target-bona-fide comparison, and a 264-pair recording- and speaker-disjoint replication. Auditing both error rates at a preset threshold is standard biometric practice [19, 16, 17], and [20] holds a cost-optimal threshold fixed across the same 21LA codecs but reports relative cost only; what remains open is how a threshold set on one channel behaves on another when only bona fide is available to reset it, which is what we measure. ReplayDF reports channel-induced spoof-side degradation with little change in bona-fide acceptance [21]; we measure transported against target-calibrated thresholds at a fixed bona-fide error target.

Calibration is also well established in speech evaluation. Challenges audit fixed-cost operating points through t-DCF/a-DCF [16]; and speaker recognition has long used C_{lr} and application-independent calibration measures [22]. These methods use both classes, which are unavailable in our setting, and threshold calibration here is distinct from the probability calibration evaluated with ECE in [23]. We study general countermeasure scores, specify a finite-sample estimand, and compare policies at the same labeling budget.

Prior work detects deployment drift and then retrains the model [24]; we hold the detector fixed and ask whether its threshold remains controlled. Distributional distance alone does not certify an FPR band, and none of our monitors survives an attack-prevalence shift. In a separate pre-specified test, replacing 25 of the 500 cohort recordings with spoofs pushes realized FPR below target on seven of eight cells (0.31–4.99%); the eighth, overlap-dominated cell is nearly unchanged at 5.16%. Non-exchangeable conformal theory does not determine the direction of error [25]. Covariate-shift weighting [26], outcome-adaptive conformal inference [27], and contamination-robust methods [28] motivate distinct remedies, but we test only the score-weighting heuristic defined in §4. The negative result is therefore not attributed to those constructions. ASVspoof 5 [17] audits a preset threshold in-domain, but its cross-dataset tables remain EER-only.

3. THRESHOLD POLICIES

We treat spoof as the positive class. A detector emits score s (higher = bona fide), and deployment flags spoof when $s < t$, so $\text{FPR}(t) = \Pr(s < t \mid \text{bona fide})$ is the bona-fide rejection rate ($P_{\text{miss}}^{\text{cm}}$ in the ASVspoof convention) and $\text{FNR}(t) = \Pr(s \geq t \mid \text{spoof})$ the spoof acceptance rate. “Calibration” below means selecting t for a target FPR, not probability calibration. We compare policies by resource class. With **no target data**, the source dev threshold t at the target FPR is transferred unchanged. With **unlabeled target data**, we fit the corrections of [7] to target statistics; we reimplement C1 z-norm, C2 temperature/shift, and C5 AS-norm. With N **labeled target bona-fide recordings**, we test three policies: (a) cohort z-norm standardizes scores by the cohort mean and standard deviation, then applies the source threshold in standardized units; (b) the Gaussian parametric quantile sets $t = \bar{s}_N - 1.645 \hat{\sigma}_N$; and (c) the *conformal quantile* sets t to the k -th smallest cohort score, where $k = \lfloor (N+1)\alpha \rfloor$. Exchangeability without ties gives the last policy’s marginal rank guarantee. Under iid continuous sampling from a fixed score distribution, its conditional FPR across calibration draws additionally follows $\text{Beta}(k, N+1-k)$ [12, 13]. The **target-bona-fide oracle** sets the threshold realizing α on deployment bona fide; spoof labels measure its FNR but do not select it. All labeled policies consume identical paired cohort draws. **Definitions.** A detector–corpus cell is *overlap-dominated* when its oracle FNR at the target FPR exceeds 50%: it has no usable operating point under our joint criterion, is italicised in Table 1, and is excluded from claims about usable cells. Mean-control claims average calibration draws. The $\text{Beta}(k, N+1-k)$ dispersion is an iid-continuous population reference, not the exact law of finite-pool draws without replacement; we report the held-out empirical spread separately over $B=1000$ draws. C4 (CORAL) of [7] requires embeddings through the frozen head, which is a different resource class, so C2 substitutes.

4. EXPERIMENTS

Setup. We evaluate AASIST [29], SSL-AASIST [5], and XLS-R+SLS [6]. The first two use released 19LA-trained checkpoints; where official XLS-R+SLS scores are available, our scores agree at per-trial $r=0.994$. The corpora are ASVspoof 2021 LA and DF [2], In-the-Wild [3], BRSSpeech-DF [4], and ASVspoof 5 eval [17] for replication. The 2021 keys carry *eval*, *progress* and *hidden* phases; the hidden phase removes non-speech by voice-activity detection and was not used in challenge results [30]. Non-speech is a documented shortcut for these detectors [20, 31], so the trimmed subset is a shortcut-removed benchmark [32] rather than a deployment channel, and mixing it into a calibration cohort mixes two score populations. A first version of this study did so; as a post-hoc protocol correction, every result below excludes

it and keeps the two untrimmed phases, and restricting to the eval phase alone changes no count reported below. Each 21LA condition is then the same 2,356 bona-fide recordings of 67 speakers and 21,164 spoofs: six conditions are real transmissions, five through an Asterisk PBX over VoIP and one over a PSTN carrier, and one is the untransmitted source [2]. By contrast, 21DF is compression-only and untransmitted; the reproduction-scored detectors use a random 100k-trial subsample of it (97,051 trials, 3,430 bona fide, after the exclusion), while official XLS-R+SLS scores are evaluated on the full 21LA and 21DF sets (16,492 and 20,637 bona fide). ASVspoof 5 has no hidden phase. All thresholds target $\alpha=5\%$. Labeled policies use the same paired cohort draws ($B=1000$), and FPR is measured on held-out data. Budget sweeps cover $N=30-10^4$; full-grid comparisons use $N=500$.

FPR control and FNR cost. At $N=500$, the conformal quantile’s mean realized FPR is 4.91–5.06% across the 12 detector–corpus cells (Table 1), and the 2.5–97.5 percentile spread of realized FPR over the draws lies within 2.7–7.6% on every cell, close to the iid Beta reference. On every cell with a usable operating point, the quantile’s FNR premium over the oracle threshold is ≤ 0.5 points. Across budgets on ITW with SSL-AASIST, cohort z-norm stays 7–8 pp high and the Gaussian quantile 0.6–1.4 pp high.

Matched-resource policy comparison. With SSL-AASIST, $N=500$, and 1000 paired draws, the conformal quantile realizes 5.0% FPR on all four corpora, as expected from marginal rank validity. The informative comparison is how the alternatives behave with the same resources. Naive transfer misses target by 17–93 pp, and the best unlabeled correction misses by 3–95 pp. C5 AS-norm is excluded as degenerate because it controls FPR only by driving FNR to $\approx 99\%$. The Gaussian parametric quantile is the fair competitor at the same N because it also targets a target-domain FPR. It misses by up to 7.0 pp (0.68 pp on ITW) and collapses to 0% on BRSpeech. Cohort z-norm does not target a target-domain FPR, so its larger errors (2–18 pp, 6/8 cells beyond ± 2 pp, and up to 50.8 pp for XLS-R+SLS) are a category difference rather than a direct contest.

Drift is detector-conditioned (Fig. 1). We measure severity as $\log_2(\text{FPR}/\alpha)$. An additive band has only α points below target and therefore cannot distinguish a collapsed threshold from a controlled one. Over the 108 cells, 77 miss target by more than $2\times$ (57 of the 84 within-corpus cells). On the ratio scale, 41 of the 72 cells inside our own ± 5 pp tolerance on realized FPR miss target by over $2\times$, necessarily conservatively, since that tolerance excludes a liberal $2\times$ miss by construction. The $2\times$ bar is post-hoc, and the count depends on that choice: it ranges from 54 of 72 at $1.5\times$ to 29 of 72 at $4\times$. We therefore attach no interval. 58 of those 72 cells are within-corpus, where calibration and deployment are the same 2,356 recordings from the same 67 speakers under seven 21LA channel conditions, so they are

not independent observations and no significance claim is made of them; the replicate below tests whether the finding survives this dependence. Their spoof-side cost is measured against the oracle threshold for that deployment, not against the calibration condition, and reaches +0.69 (AASIST, PSTN→G.722, whose 0.42% FPR is 10 of 2,356 bona fide; two cells score marginally higher but their FPR rests on fewer than five expected false alarms, $\text{FPR} < 5/n$, and is censored). This cell is not isolated: every AASIST cell calibrated on PSTN pays at least +0.40, and on PSTN itself the same threshold misses 31% of spoofs. PSTN is AASIST’s least separable 21LA condition (AUC 0.956 against 0.975–0.997 elsewhere), so a threshold set there sits below the bona-fide scores of every cleaner channel. Per calibration draw a positive spoof-side cost places the threshold below oracle and so inside the ± 5 pp tolerance; empirically every positive-price cell passes it, which is why the additive test cannot reveal this cost. AASIST pays the larger spoof-side cost. **Replication on recording-disjoint data.** Because the grid above reorders one 2,356-recording set, we repeat the experiment on ASVspoof 5 eval [17], where calibration and deployment share neither recording nor speaker. The grid contains 11 organizer-applied codecs plus the unprocessed source and 737 speakers. It is restricted to the 80.5% of bona-fide sources appearing under one condition and split 50/50 by speaker. **Over both detectors the transported threshold misses target by over $2\times$ on 187 of 264 pairs (71%), against 68% on the grid above:** the effect persists without shared recordings or speakers; the numerical rate difference is confounded by corpus, codecs and attacks. This is descriptive evidence from one corpus, not a correction factor. **The spoof-side replication is narrower.** AASIST is overlap-dominated on this corpus (oracle FNR 63.9% against SSL-AASIST’s 7.9%), so our own cutoff excludes it, and the flagship cell above is an AASIST cell. The mechanism replicates; that cell does not; the spoof-side axis here rests on one detector. On three further detectors with released 21LA scores (XLS-R+SLS, XLSR-Mamba, XLSR-Conformer), the transported threshold misses target by more than $2\times$ on 25–33 of the 42 channel pairs each (88 of 126), while their spoof-side cost on PSTN→G.722 stays below 3 points: the FPR-axis failure is detector-general, the spoof-side magnitude is not.

Monitoring was pre-specified as a co-primary negative-result analysis with acceptance $\text{TPR} \geq 0.8$ and $\text{FPR} \leq 0.2$ for flagging drifted cells. The flag definition was changed after results existed, from “realized FPR more than 2 pp off target” to $|\log_2(\text{FPR}/\alpha)| > 1$; under the pre-specified definition no monitor has an operating point. Under the corrected definition the mixture- W_1 monitor, the W_1 distance between deployment and calibration mixture-score distributions, has an in-sample operating point on SSL-AASIST only (threshold selected on the same cells), but re-mixing the deployment spoof share by $\pm 50\%$ pushes 19/19 and 18/19 benign cells over it. The entropy monitor (mean binary entropy of

Table 1. Upper block: mean FNR (%) over $B=1000$ paired $N=500$ cohorts at the conformal quantile targeting $\text{FPR}=5\%$; parentheses give mean FNR minus deployment-oracle FNR, in points. The pooled 21LA EER spans the seven conditions: per condition it is 1.9–11.1 (AASIST), 0.2–1.0 (SSL-AASIST) and 0.5–3.5 (XLS-R+SLS). Italics: overlap-dominated cells (oracle FNR $>50\%$), where FPR control holds but no usable operating point exists. Analysis code, score tables and result files: <https://github.com/rvirgilli/speech-deepfake-threshold-transport/tree/a2-icassp2027-final>.

System	21LA	21DF	ITW	BRSpeech
AASIST	19.1 (+0.4)	36.8 (+0.3)	95.2 (0.0)	84.0 (−0.2)
SSL-AASIST	0.2 (+0.0)	7.1 (+0.2)	18.4 (+0.3)	98.5 (0.0)
XLS-R+SLS [†]	1.5 (+0.1)	0.7 (+0.1)	11.3 (+0.5)	91.5 (+0.4)
<i>EER (%) on the same trials</i>				
AASIST	8.3	14.5	43.0	36.1
SSL-AASIST	0.8	5.9	11.2	34.7
XLS-R+SLS [†]	2.8	2.0	7.5	34.2

[†]Official author-released scores on the full 21LA and 21DF sets.

sigmoid-standardized deployment scores) fails on both detectors under either definition. **Label-free heuristic.** We test an importance-weighted quantile that reweights the bona-fide cohort by a clipped ratio of deployment to calibration *mixture-score* densities (15 equal-width bins over the pooled score range, ratio clipped to $[0.1, 10]$, 200 draws). It holds 17/108 cells within 2 pp of target, compared with 20/108 unweighted, and restores a deployable operating point on neither detector. This is a heuristic rather than the construction of [26], which requires a target/source bona-fide covariate ratio and includes the test-point weight; our negative result is limited to the heuristic. Adaptive conformal inference [27] needs the label of each flagged item; the unlabeled mixture flag rate alone does not identify FPR when attack prevalence and the spoof-side detection rate are unknown, and we evaluate no estimator that adds identifying assumptions.

5. DISCUSSION

Calibration and detection are distinct. Target-channel calibration controls the *marginal mean*, not every realized threshold. Under iid continuous sampling from a fixed distribution, a single $N=100$ draw has only 65.5% probability of landing within ± 2 FPR points; at $N=500$, the probability is 96.2%, with a 3.26–7.06% central 95% interval. Calibration also cannot repair poor discrimination: on overlap-dominated cells, no threshold meets our joint criterion of $\text{FPR} \leq 5\%$ and $\text{FNR} \leq 50\%$ [7]. **Operational implications.** Report FNR at controlled FPR alongside EER, choose N from the dispersion required by the deployment (under iid sampling, $N=450$ gives 95.4% probability of landing within ± 2 points), ship

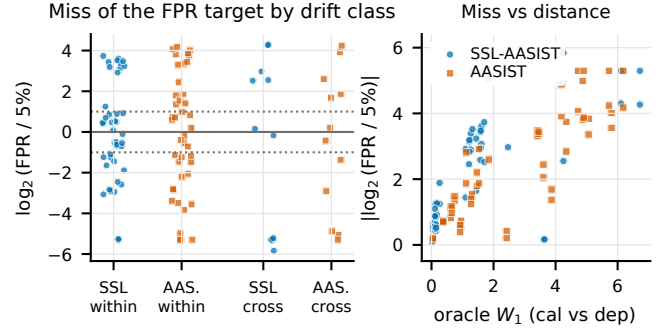


Fig. 1. Drift map. Left: signed $\log_2(\text{FPR}/5\%)$ per deployment cell (within-21LA channel pairs vs cross-corpus; colour/marker = detector). Under a post-hoc $2 \times$ two-sided bar, 41 of the 72 cells inside the ± 5 pp tolerance are off target (54 at $1.5\times$, 29 at $4\times$; no interval, §4), every one conservative. The costliest on the spoof side are among them: AASIST PSTN $\rightarrow$ G.722 misses 70% of spoofs against 0.66% for that deployment’s oracle while reading 0.42% FPR. Right: the miss correlates with oracle bona-fide W_1 (ρ 0.74/0.84) but a line fitted on one detector explains little of the other (held-out R^2 0.53/0.25), so no predictive law is claimed from it.

a designated bona-fide calibration split, and recalibrate after a channel change. **Limitations.** The Beta dispersion law assumes iid continuous recordings from a fixed deployment speaker mixture. A post-hoc speaker-disjoint diagnostic on 21LA, which calibrates on whole speakers until the cohort holds at least 500 recordings (median 15 speakers) and evaluates on the rest, widened the 5th–95th percentile spread of realized FPR on all six tested detector–condition cells to 6.0–10.1 points (means over ten seeds of 400 draws), against 3.4–3.5 under a speaker-permutation null that preserves group sizes (200 permutations); deployment to new speakers therefore requires speaker-level validation. Conclusions rest on three detectors, the drift map and the ASVspoof 5 replicate on two; the language-shift regime rests on one corpus ($n=1$); and bona-fide labeling is assumed cheap relative to spoof labeling, which holds where a deployment can verify its own recordings’ provenance. Price and severity are two readouts of threshold displacement. Their strong within-grid rank association does not make spoof-side cost identifiable from bona-fide scores alone.

6. CONCLUSION

Threshold transport is easy to mis-audit: conservative collapse makes FPR read below target while missed spoofs multiply. A bona-fide order statistic from the target channel restores marginal expected control, but dispersion depends on sample size and sampling unit, and cannot repair class overlap. Recording- and speaker-disjoint data also exhibit transport failure; the label-free heuristic tested does not restore control.

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